Capability Emergence Can Be Forecast:
Per-Seed, In Advance, With Calibrated Intervals, Certified False Alarms, and a Blind Pre-Registered Gate

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Abstract

Emergent capabilities are widely treated as unpredictable: aggregate loss improves smoothly while specific abilities appear abruptly. Prior work offers early-warning *indicators* but never scores them as *forecasts* — no lead time at a controlled false-alarm rate, no calibration, no negative controls, no blind validation. We supply that discipline and show that, in controlled settings spanning grokking model systems and small language models, emergence *timing* is forecastable per training run, in advance, with calibrated uncertainty. On language models: across 30 transformers at identical configuration — where the strongest published baseline predicts a single constant [1] — the formation time of the previous-token head (the mechanistic precursor of induction [2]) forecasts each seed's induction emergence at Spearman $\rho = 0.977$ (95% CI [0.911, 0.995]) with median lead 975 steps (∼15% of training). A best-case training-loss rule *ties* that ranking ($\rho = 0.977$) but with 50-step median lead — a nowcast, not a forecast. Split-conformal intervals of width 300 steps, anchored at the precursor alarm, covered 15/15 held-out seeds at 90% nominal. The frozen rule then passed blind pre-registered gates on *two* never-seen configurations — a wider model with sparser repetition (10/10 interval coverage, bar 7/10) and an entirely different language (9/10, exactly nominal) — with zero false alarms throughout. A trap-language rung then attacked our own rule as pre-registered: in a language where previous-token context pays for the task itself, the bare precursor false-alarms on 10/10 capability-blocked runs, while the mechanism-composed conjunction is certified in *both* language classes (0 false alarms) and times emergence at $\rho = 1.000$. Finally, a gap-origin study broke the fixed offset itself (both learning rate and batch size move the gap ∼2.3×; no external clock owns it) and revealed the scale-invariant law beneath: across 80 valid-anchor runs spanning every fleet, both languages, and all configuration axes, the anchor fires at 0.843 of time-to-emergence (IQR 0.825–0.865) — $t_{\text{event}} \approx 1.19 \times t_{\text{anchor}}$ — and this multiplicative rule passed its own blind gate at a third never-seen configuration (5/5 coverage). All false-alarm claims are certified against 28 *manufactured capability-blocked negatives* (data-ablated and architectural, across both languages) — a resource we argue is mandatory for capability monitoring and that no public suite provides. Public-checkpoint evidence: the precursor leads the capability across *three model families* (Pythia, OLMo, OLMo-2; seven suites, pre-registered rules) — OLMo-2's 1B-token checkpoint shows the precursor formed while the capability is absent — yet all Pythia suites cliff at the same token count, confirming that public artifacts cannot test per-seed forecasting. The grokking half of the paper (466 runs) contributes the evaluation methodology and its hard lessons: a robustness-versus-false-alarm tradeoff, mechanism-factored gates that resolve it, non-transfer of fitted calibration, and a label-free probe family that died under pre-registered confirmation. Four pre-registered kill criteria fired across the program and are reported. Every prediction, freeze, and verdict is commit-stamped before its data existed. Scope: named capabilities with known mechanistic precursors, small models; we state what stands between this and frontier deployment.

1 Introduction

When a capability emerges during training, the training curve rarely announces it in advance: loss declines smoothly while the ability arrives abruptly [2, 3]. For frontier-scale systems this unpredictability is a safety problem with a name — labs want to know what a model will be able to do *before* it can do it. A literature of early-warning signals has grown in response: loss-curve oscillations that predict whether grokking will occur [4], spectral observables that show precursors of phase transitions [5], geometric stage detection [6], and scale-axis extrapolations [7, 8]. What none of this work does is score a *forecast*: state, in advance, when a capability will arrive; attach calibrated uncertainty; measure lead time at a controlled false-alarm rate against negatives where the capability never arrives; and validate the frozen system blind. Indicators without that discipline are weather lore, not weather forecasting.

This paper does two things. **First**, it builds the missing evaluation discipline on a grokking model system — 466 training runs across two domains with interventions that shift emergence timing ∼100× — and reports what the discipline kills: single-signal alarms die on look-alike negatives; the context features that control false alarms are exactly the features that break under interventional shift (a robustness–false-alarm tradeoff); fitted calibration does not transfer across regimes; and an initially spectacular label-free probe family failed its pre-registered confirmation. **Second**, it carries the surviving methodology to language models and obtains the result the discipline was built for (Figure 1): per-seed forecasts of induction-head emergence from a mechanistic precursor, with calibrated intervals, certified false alarms, and a blind pre-registered transfer gate passed at ceiling.

Contributions, each with its adversarial control:

1. **A forecasting benchmark and protocol for emergence timing**: event definitions robust to outliers, lead time at capped false-alarm rate, champion selection on training data only, (correlation, lead) reported as a pair — plus a *negative taxonomy* (budget-censored vs. structurally blocked vs. *manufactured*) and a minimum-negative-count rule learned the hard way.

2. **The per-seed result** (Sections 5.2–5.3): at fixed configuration, where the config-equation baseline of Aoyama and Wilcox [1] is a constant by construction, precursor formation time forecasts emergence at $\rho = 0.977$ with 975-step median lead; a best-case loss rule ties the correlation with 50 steps of lead. Warning time lives in *where you anchor*, and only the circuit precursor anchors early.

3. **Manufactured negatives, and the trap that proves they matter** (Sections 5.2, 5.5): capability-blocked training runs — data-ablated and architectural — enabling the first certified false-alarm rates for emergence forecasting. In simple languages the bare precursor is certified (0/10 FA); in a trap language built so previous-token context pays for the task itself, it fails completely (10/10 FA on blocked runs, as pre-registered) and the mechanism-composed conjunction is certified in both regimes (0 FA) while timing emergence at $\rho = 1.000$. Anchors must be composed from mechanism, not read off a single circuit.

4. **Three blind gates and a scale-invariant law** (Sections 5.4, 5.6): the frozen rule passed pre-registered blind gates on two never-seen configurations (10/10 and 9/10 coverage); a gap-origin study then broke the fixed offset (both optimization axes move the gap; no external clock owns it) and exposed the law beneath — the anchor fires at 0.843 of time-to-emergence across 80 runs — whose multiplicative form passed a third blind gate (5/5) at yet another unseen configuration.

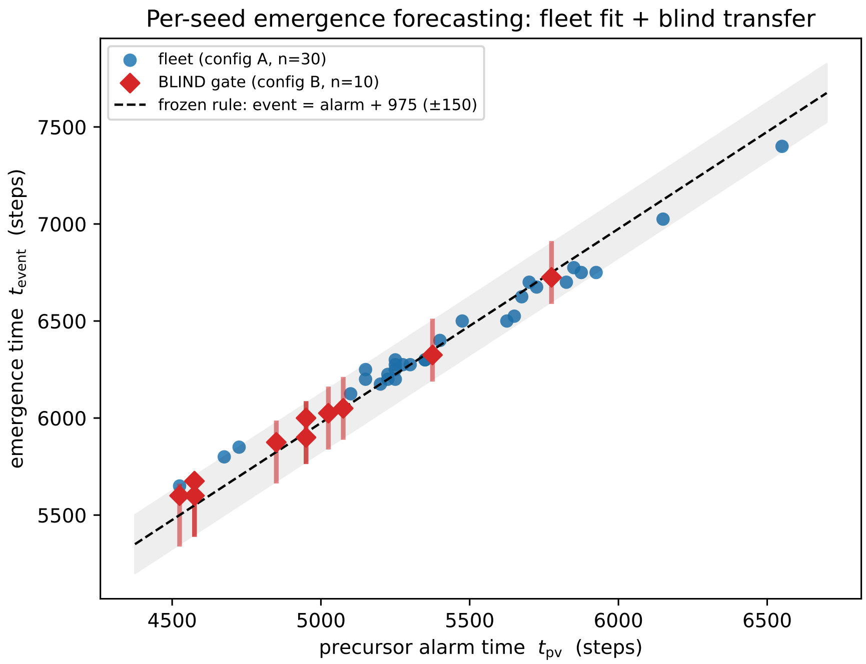

Figure 1: **The central result.** Precursor alarm time vs. emergence time. Blue: the calibration fleet (30 seeds, config A) — Spearman $\rho = 0.977$. The dashed line and band are the *frozen* rule (event = alarm $+975 \pm 150$). Red diamonds with bars: the blind gate — 10 fresh seeds on a never-seen configuration (wider model, sparser repetition), scored once against the frozen rule: 10/10 covered. Negatives (18 manufactured capability-blocked runs, not shown) produced zero false alarms.

5. **The kill ledger** (Section 6): four pre-registered kill criteria fired across the program and are reported with their mechanisms, alongside every failed prediction. The positive results above survived the same regime that produced these deaths; that, not any single number, is the paper's claim to trust.

Everything is pre-registered in a public commit chain (every freeze precedes its data; Section 9), and the corpus, fleets, and harness are released as a benchmark.

2 Related work

Notsawo et al. [4] predict *whether* grokking will occur from early loss-curve oscillations — a scalar whether-classifier; its oscillation feature is one of our baselines. Anonymous et al. [5] derive spectral early-warning indicators (2-RDM heat capacity) across four transition settings but report no lead time, calibration, false-alarm rate, or baseline race; our label-free rung (Section 6) shows concretely why unscored spectral indicators can mislead: ours died under confirmation. Hoogland et al. [6] detect developmental stages retrospectively via the local learning coefficient. Aoyama and Wilcox [1] predict induction-head formation time from pre-training config (batch, context, bigram statistics) — powerful, but constant per config; our fleet result answers precisely the variance their equation cannot (per-seed timing at fixed config), and our Pythia analysis confirms their account across public suites (all five cliff at the same token count). Scale-axis emergence prediction [7, 8] forecasts across model size, orthogonal to our within-run time axis. Olsson et al. [2] established the induction-head phase change and its previous-token precursor; Barak et al. [9] and progress measures [10] established hidden progress; we turn these observations into scored forecasts. The grokking substrate builds on [11–13] and our companion papers [14–16], which established the representational-timing mechanism and the norm clock that our two-gate forecaster exploits.

3 Forecasting emergence: definitions and discipline

A **run** logs behavioral and internal signals at fixed cadence. An **event** is the first crossing of an absolute behavioral criterion robust to outliers (never a fraction of a run's own maximum — a rule adopted after that operationalization failed twice; Section 6). A **forecaster** observes the log prefix and may **alarm** once; its **lead** is $t_{\text{event}} - t_{\text{alarm}}$ (alarms after the event score zero; no alarm on an eventing run is a miss). **False-alarm (FA) rate** is the fraction of *negative* runs — runs where the capability never arrives — that alarm. Thresholds and champions are selected on training runs only, under an FA cap ($\leq 5\%$), with at least five negatives on every side of every split (a minimum learned from a degenerate fit, Section 6). Negatives come in three kinds: *budget-censored* (would event with more budget — these legitimately attract alarms), *structurally blocked* (wrong learned structure), and *manufactured* (constructed so the capability cannot form). Forecast quality is always a *pair* — (ranking skill, lead) — because a rule can rank perfectly by detecting the event as it happens; and point forecasts carry split-conformal intervals whose empirical coverage is reported. Blind validation means: every constant frozen and commit-stamped, then fresh runs, one scoring pass.

4 Part I: the grokking benchmark, and what the discipline kills

We built the protocol on 466 pre-existing grokking runs (modular arithmetic transformers; MNIST MLPs in the Omnigrok regime) whose companion-paper interventions (contrastive priors, norm clamps) shift emergence timing up to $\sim$100$\times$ and — crucially — *decouple* the dominant scalar clock (weight norm) from the outcome. Full protocol and verdicts are in the repository (`plan_p5.md`, `P5_RESULTS.md`); four results shape everything downstream.

(i) In-regime forecasting works. A multivariate forecaster with mechanism-motivated features achieved 10,400-step median lead (60% of the delay) on held-out seeds at zero structural false alarms, and 18,000 steps prospectively on pre-registered fresh runs including a never-seen norm pin (bar 5,200) — all four prospective predictions passed.

(ii) The robustness–false-alarm tradeoff (Figure 2). Trained on control runs only and tested on intervention arms, the scalar norm clock collapses (median lead 0; the interventions hold the norm where controls never generalize), and the FA-controlled multivariate collapses with it — *because* the context features that suppress false alarms are the features the intervention breaks. The bare content probe survives (4,400–16,400 steps). A monitor tuned quiet on its training distribution goes blind exactly when the recipe changes.

(iii) Mechanism-factored gates resolve the tradeoff where mechanism is known. Composing a content gate with an *a-priori* norm-viability window (constants from the companion papers, not fit) retains 12,400 steps (53% of the delay) under the same shift at zero miss. The portable ingredient is mechanism, not fitted calibration: fitted thresholds transferred across domains in neither direction (kill K5).

(iv) The discipline kills seductive numbers. An algorithmic-domain corpus with structure-complete-but-blocked negatives admitted *no* forecaster at 5% FA (kill K1/K8): trap negatives carry more "progress" signal than true positives show before their events. And a label-free spectral probe

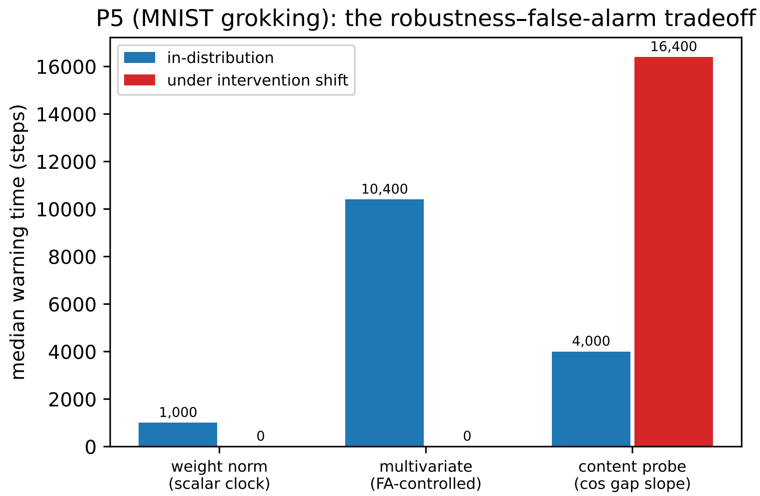

Figure 2: **The robustness–false-alarm tradeoff** (MNIST grokking; median warning time at $\leq 5\%$ FA, higher is better). The scalar clock (weight norm) forecasts in-distribution but collapses when interventions decouple the norm from the outcome. The multivariate forecaster that best controls false alarms in-distribution (10,400 steps, 60% of the delay) collapses hardest under shift — its norm-context features are exactly what the intervention breaks. The bare content probe survives the shift (16,400 steps) but is FA-fragile in corpora with look-alike negatives.

that posted a 34,200-step validated lead died under pre-registered confirmation — it false-alarmed on fresh structural negatives and, per-run, was largely detecting the *intervention*, not the emergence (kill K9). Both deaths are in the ledger (Section 6) and both shaped the LM protocol.

5 Part II: language models

5.1 Public checkpoints: the precursor leads, and why public suites are not enough

We probed 5 Pythia suites [17] (70m/160m/410m, standard and deduped) across training checkpoints for three quantities: behavioral copy advantage on repeated random sequences, the maximum per-head prefix-matching (induction) score, and the maximum early-layer previous-token score — the known mechanistic precursor [2]. The phase change is crisp in every suite (Figure 3): at 1.07B tokens copy advantage is 0.022 and the precursor is already at 0.37 ($\sim$10$\times$ baseline) while induction sits at 0.019; one stage later both cliff (copy 9.7 nats, induction 0.93). Against a best-case loss threshold granted an oracle sweep, the precursor alarmed strictly earlier on 4/5 suites and never post-event (leads 1–8 stages); our pre-registered prediction here failed in both directions — the precursor led by *more* than predicted and loss was *stronger* than predicted (one stage of genuine timing) — and is reported as registered. Two structural facts matter more than the scores. First, all five suites, which share batch and context configuration, cliff at the same $\sim$2.1B tokens — consistent with config-determined timing [1] and fatal to using public suites for per-seed forecasting: there is one run per config. Second, an event definition we had frozen (50% of each run's maximum) was silently broken by a late-training outlier in one suite — the same operationalization failure our grokking rungs had already punished — and was replaced, in print, by absolute criteria everywhere.

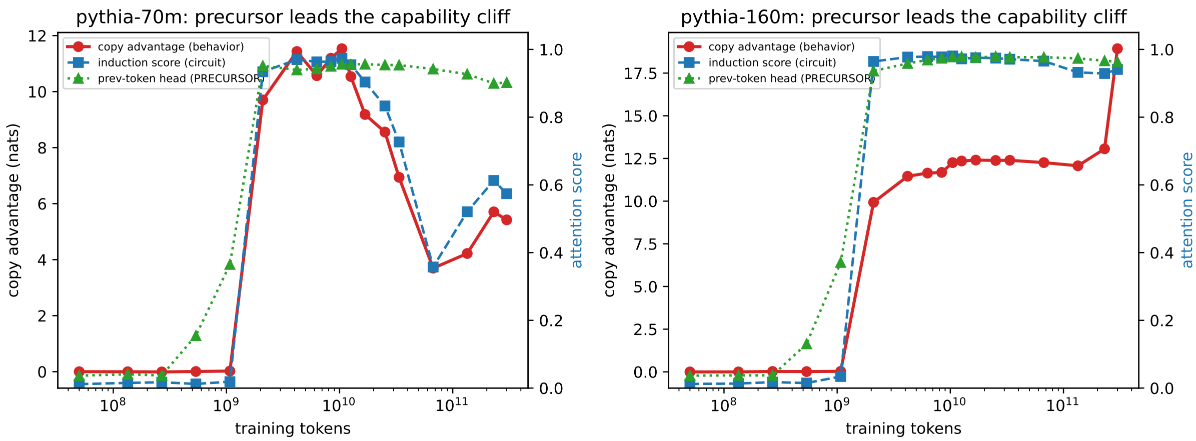

Figure 3: **Public Pythia checkpoints (models we did not train).** The previous-token head (green) rises a full public-checkpoint stage before the induction-head score (blue) and the behavioral copy advantage (red) cliff together. Left: pythia-70m; right: pythia-160m. The 70m model also partially *loses* induction late in training (prefix 0.93→0.36 at 67B tokens) — capability regression exists and is monitorable.

Cross-family replication (pre-registered). To test whether the precursor-leads phenomenon is a Pythia artifact, we froze the same rules (precursor ≥ 0.10; capability ≥ 2.0 nats) and swept two families this program had never probed — AllenAI's two OLMo generations, with architectures, data pipelines, and tokenizers all distinct from Pythia's and from each other. All three pre-registered predictions passed in *both* families, and each family's public grid caught the intermediate state directly: OLMo-1 at 8B tokens shows the precursor formed (0.43) with the capability below threshold (0.71 nats), and OLMo-2 at 1B tokens shows the precursor formed (0.22) with the capability *absent* (-0.14 nats) — ordinal leads of 8→12B and 1→21B tokens respectively. The kill (capability before precursor in any family) fired nowhere. Across three model families and seven public suites, the precursor leads.

5.2 The seed fleet: per-seed forecasts, certified against manufactured negatives

Public suites cannot answer the forecasting question; fleets can. We trained 45 two-layer transformers on a fixed synthetic language (bigram-structured, with variable-offset within-context repetition so that positional shortcuts cannot substitute for induction — a shortcut our first design permitted, caught in smoke testing): 30 positives at one configuration (seeds vary; config fixed), plus 15 **manufactured negatives** — 10 *data-ablated* (identical language, zero repetition: induction has no training signal) and 5 *architectural* (one layer: induction is structurally impossible [2]). Events use an absolute criterion (copy advantage ≥ 2 nats, two consecutive evals).

Results, scored once against the pre-registered spec. Per-seed event times span 5,650–7,400 steps (spread 1.31×; the pre-registered bar of 1.3× was cleared by 0.01 — variance at fixed config is real but modest, and we do not dramatize it). The precursor's crossing time forecasts each seed's event: $\rho = 0.977$ (bootstrap CI [0.911, 0.995], $n = 30$), median lead 975 steps (825–1,125) — ∼15% of the run issued as advance notice, seed by seed, where the config equation predicts one constant. The best-case loss threshold ($\theta = 2.077$, granted an oracle sweep) *ties* the ranking ($\rho = 0.977$) — and forecasts nothing: its median lead is 50 steps (minimum -25), i.e., it detects the cliff in progress. Our pre-registered comparison metric (rank correlation alone) was therefore *wrong as designed*; the two-dimensional truth — equal correlation, 20× the lead — is reported alongside the registered failure, and (correlation, lead) pairs are now the protocol's required form. A second dissociation: the graded *behavioral* ramp that precedes the cliff carries no timing information ($\rho = -0.03$) — circuits time the transition; early behavior does not.

The negatives certified the false-alarm side. On data-ablated runs the precursor never formed (max layer-0 previous-token score 0.036–0.061 vs. threshold 0.10): bare precursor FA 0/10, conjunction FA 0/10, with pre-event alarms on 30/30 positives. Here our pre-registered mechanistic prediction was wrong and is reported: we expected the precursor to form anyway (predicting false alarms that would require a conjunctive gate); in fact a pure bigram language is solvable through the embedding pathway alone, so previous-token attention only pays where repetition exists. The manufactured negatives thus *certified* the precursor rather than trapping it — certification is the other thing negative sets are for — and constructing a language where the trap is real (where previous-token context pays independently of repetition) is the named follow-up. Architectural negatives: zero events, induction score ≤ 0.014 throughout, zero alarms.

5.3 Calibrated intervals

Single thresholds are fragile (a P5 lesson bought with a dead result); forecasts should be intervals with measured coverage. Split-conformal calibration (seeds 1–15) around the anchored point forecast $t_{\text{pv}} + 975$ yields intervals of median width 300 steps that covered 15/15 held-out seeds (nominal 90%) — issued 975 steps before the event. The identical machinery anchored at the loss crossing covers too — with 50 steps of lead. Anchoring, not calibration, is where warning time comes from.

5.4 Two blind gates

Everything above is one configuration. The ship-gate: freeze the entire rule verbatim — alarm at layer-0 previous-token score ≥ 0.10; forecast the event in $[t_{\text{alarm}} + 825,\ t_{\text{alarm}} + 1125]$ — commit it, then score once on fresh runs the rule never saw. **Gate B** (model width 256 → 320; repetition density 0.75 → 0.6; 10 seeds + 3 negatives): 10/10 events, 10/10 pre-event alarms, **10/10 interval coverage** (bar 7/10), $\rho = 0.988$, 0/3 false alarms, median lead 1,012 steps. **Gate C** (the language itself replaced — a different generator seed produces a different bigram table; original architecture): 9/10 coverage — *exactly* the 90% nominal the conformal quantile promises — 10/10 pre-event, 0/3 false alarms (bare rule also 0/3), $\rho = 0.952$, median lead 987.5. Our pre-registered expectation was humble both times: the P5 transfer kill made the frozen offset the component we flagged as most at risk. It survived both shifts — and Section 5.6 explains why, less flatteringly than we had hoped.

5.5 The trap language: our rule, attacked as pre-registered

The fleet's negatives certified the bare precursor partly because a pure bigram language gives previous-token attention no task role (Section 5.2). So we built the language where it has one — next-token distributions selected by the previous token (one learnable bit of pair context) — and pre-registered the attack on our own headline rule: the precursor should now form *everywhere*, including where induction cannot. It does. On 10 capability-blocked runs in the trap language the bare precursor rule false-alarms **10/10** (crossing threshold at 650–4,400 steps from the task incentive alone), and on 2/10 positives the graded behavioral signal false-alarms transiently on its own (2/10). Neither gate survives alone. The *same-eval conjunction* of the two — frozen a priori in the pre-registration — is certified in both language classes (0/10 FA here; 0/10 in the bigram fleet) and, remarkably, times emergence *better* than the bare precursor ever did: $\rho = 1.000$ across 10 seeds with median lead 1,900 steps (range 1,525–2,100). A secondary anchor — the induction

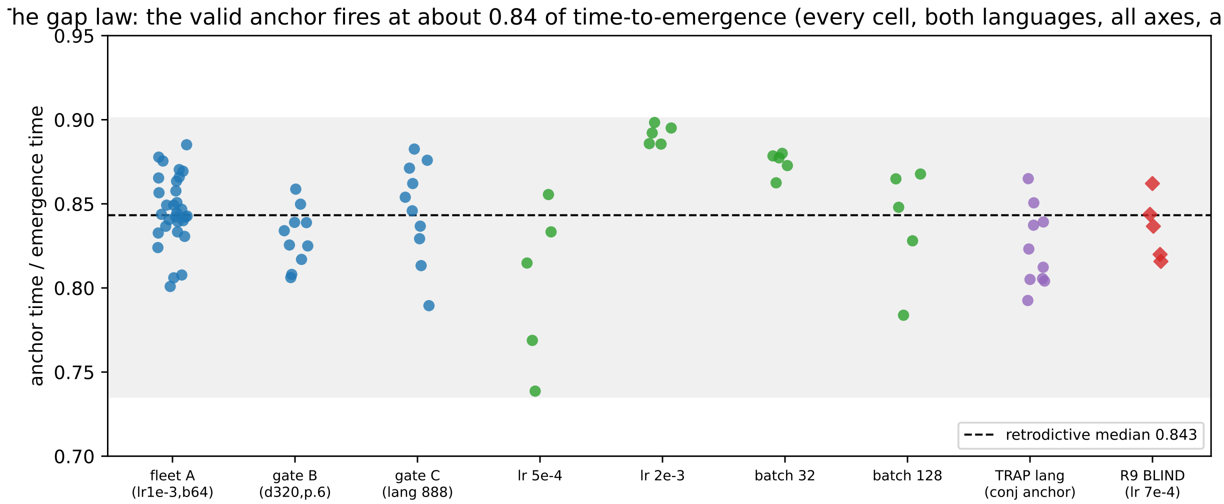

Figure 4: **The gap law.** Anchor time over emergence time for every valid-anchor cell in the program: the calibration fleet, both blind gates, all four gap-probe cells (learning rate halved/doubled, batch halved/doubled), the trap language via its conjunction anchor, and — red diamonds — the blind validation at a never-seen learning rate. Median 0.843 (dashed); the shaded band is the frozen envelope [1/1.36, 1/1.11] used in the blind test. One fraction describes 80 runs whose emergence times span 3×.

score's own early ramp — gives $\rho = 0.988$ at 550-step lead, also 0 FA. Design iterations disclosed: two earlier trap constructions failed instructively (an unlearnable language gives the precursor no incentive at all; too rich a language starves induction past budget), and both feed Section 5.6's account.

5.6 From offset to law: the gap is a fraction, not a clock

Why did a fixed 975-step offset survive two config shifts? The pre-registered gap-origin study answers by breaking it. Halving the learning rate stretches the gap 2.27×; halving the batch stretches it 2.35×: both optimization axes move it, so *no external clock owns the gap* (the registered discrimination rule finds no owner; kill K8b fires as written). The post-hoc analysis — labeled as such — finds the invariant underneath (Figure 4): across all 80 valid-anchor runs in the program, spanning both languages, every configuration axis, and a 3× range of emergence times, the anchor fires at 0.843 of time-to-emergence (IQR 0.825–0.865, range 0.739–0.898). The gates of Section 5.4 passed because their shifts happened to preserve the emergence timescale; the R8 cells, which change it, would have broken the fixed offset — and the scale-invariant form $t_{\text{event}} \approx 1.19 \times t_{\text{anchor}}$ is the rule that survives both. Per our discipline, an 80-run retrodictive law is exploratory until tested: we froze the envelope $[1.11 \times t_{\text{pv}},\ 1.36 \times t_{\text{pv}}]$, pre-registered it, and ran a third gate at a never-seen learning rate. Outcome: **5/5 coverage** (observed multipliers 1.160–1.226 against a frozen median of 1.19), 5/5 events, 0 false alarms on fresh negatives. The forecaster the paper ships is multiplicative; fixed-step offsets are timescale-local, and we say precisely when each applies. Why ≈0.84 — why the last sixth of the road is announced — is now the program's sharpest open question; the trap-language design iterations (capability delayed or starved by competing learnable structure, with the gap stretching in proportion) point toward a competition-for-gradient account we state as hypothesis.

6 The kill ledger

Positive results are only as credible as the regime that tried to kill them. Everything below fired as pre-registered and is reported with its mechanism:

kill / failure	where	mechanism
K1/K8: no forecaster at 5% FA	grokking (algorithmic)	trap negatives out-signal true positives
K5: calibration non-transfer	grokking (cross-domain)	fitted thresholds are regime-local
K9: label-free probe dies	grokking (confirmation)	signal was the intervention, not emergence
K8b: no external clock owns the gap	LM gap study	both axes move it; gap is a fraction of the journey
bare precursor in trap languages	LM trap rung	10/10 FA where the circuit pays for the task
fixed-offset intervals	LM gap study	timescale-local; multiplicative law replaces them
P1 (both clauses)	Pythia	precursor earlier, loss stronger than predicted
50%-of-max event rules	twice	fragile to outliers; absolute criteria adopted
P2c metric as registered	fleet	rank correlation without lead rewards nowcasts
P2d interior mechanism	fleet	the trap exists, but not in bigram languages (Sec. 5.5)
single-negative FA cap	grokking (R5)	degenerate fit; minimum-negative rule adopted

Four of these (K1/K8, K5, K9, K8b) are full kill criteria; the rest are registered predictions that failed informatively — including two rungs designed to break our own headline rule, both of which succeeded and both of which yielded their repair (the conjunction anchor; the multiplicative law). The protocol that survived this ledger is the one every language-model result was scored under.

7 Discussion: what this does and does not establish

Established. For a named capability with a known mechanistic precursor, in small transformers: emergence timing is forecastable per run, ∼15% of training in advance, with calibrated intervals (15/15, 10/10, 9/10, and 5/5 coverage across four evaluations, three of them blind), certified false-alarm rates (0 across 28 manufactured negatives spanning both language classes — for the *correctly composed* anchor; the bare anchor fails 10/10 where its circuit pays for the task), a baseline accounting showing loss curves carry ranking but no lead, and a scale-invariant forecasting law ($t_{\text{event}} \approx 1.19 \times t_{\text{anchor}}$, 80 runs, blind-validated) that replaces timescale-local fixed offsets. The evaluation discipline — manufactured negatives including trap classes, (correlation, lead) pairs, FA caps with minimum negative counts, absolute event criteria, prereg-gated freezes — is domain-general and, we argue, the minimum standard any early-warning claim should meet.

Not established. Frontier applicability. The gap has named parts: (1) *precursor knowledge* — induction heads have a known antecedent; most capabilities of concern do not, and finding precursors is mechanistic-interpretability work our benchmark can score but not replace; (2) *negatives at scale* — frontier labs have n=1 runs and no capability-blocked counterfactuals; our manufactured-negative recipe (ablate the training signal the capability needs; block the architecture) is implementable in principle at any scale, and we regard a library of blocked runs as the single most actionable transfer of this work; (3) *recipe shift* — we passed three blind gates, but the gap law says exactly when fixed calibration breaks (any shift that moves the emergence timescale), and the P5 tradeoff says shift-robustness must be re-certified per monitor; mechanism-factored designs are the only pattern we found that survives shift by construction; (4) *the fraction question* — the gap law ($t_{\text{anchor}} \approx 0.843\, t_{\text{event}}$) is blind-validated but unexplained: why the final ∼16% of the road to emergence is announced, and whether the fraction itself is architecture- or capability-dependent, is open; the competition-for-gradient account is our stated hypothesis. Emergence forecasting in the wild also faces capabilities that *regress* (pythia-70m's late induction loss) and metric-induced discontinuities [3]; our absolute-criterion and interval machinery handles both mechanically, but the monitoring literature has not.

Practical recipe (what a lab could run tomorrow, at their scale, for a named capability): identify the precursor circuit; *compose* the anchor — circuit signal conjoined with a graded behavioral signal at the same evaluation, since either alone false-alarms in some regime (Section 5.5); log both densely; manufacture blocked negatives, including a trap class where the precursor circuit pays for the task; calibrate the *multiplicative* interval on a seed fleet ($t_{\text{event}} \in [1.11, 1.36] \times t_{\text{anchor}}$ was ours; measure your own); freeze; monitor; re-certify after any recipe change that could move the emergence timescale.

8 Limitations

One capability (induction), small models (2-layer fleets; ≤1B public suites), synthetic fleet languages, three blind gates but each on one or two axes at a time. The trap prediction that harder languages reintroduce false alarms is now *tested and confirmed* (Section 5.5), with the conjunction as the repair; trap classes beyond pair-context languages remain unexplored. The gap law is blind-validated at one new configuration and post-hoc across eight; its fraction may yet vary with architecture or capability. R4's calibration seeds were not blind (the gates are). The per-seed spread cleared its pre-registered bar by 0.01. Two registered kill clauses fired against our own predictions in the LM half (K8b; the trap's interior mechanism in the bigram fleet) and are reported as registered. Pythia event times are bounded by public checkpoint granularity. All pre-registration is repository-native (commit-stamped, publicly pushed), not third-party.

9 Reproducibility

Every number in this paper is a macro generated from scored artifacts (`paper4/gen_numbers.py`, byte-verified by `paper4/verify_regen.py`). The prereg chain is public: plan and kills before code (`2cf62e3`), forecaster freezes before test blocks, fleet spec before launch (`6b05770`), each blind gate frozen verbatim before its runs existed (`377511b`; `4a5083f` for the trap, third gate, and gap study; `1e07b45` for the law's envelope), verdicts after each (`f25aa45` through `5238241`). Corpora: 466 grokking runs, 5 public-suite probe trajectories, and 118 language-model runs with dense circuit probes (calibration fleet, three gates, trap fleet, gap-probe cells, and the law's blind cell, positives and manufactured negatives throughout); training and scoring harnesses: `src/train_lm.py`, `src/probe_pythia.py`, `analysis/score_r*_p6.py`. Hardware: one RTX 3080.

References

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